

Advanced Dynare

Chapter V

Kalman Filter and the likelihood

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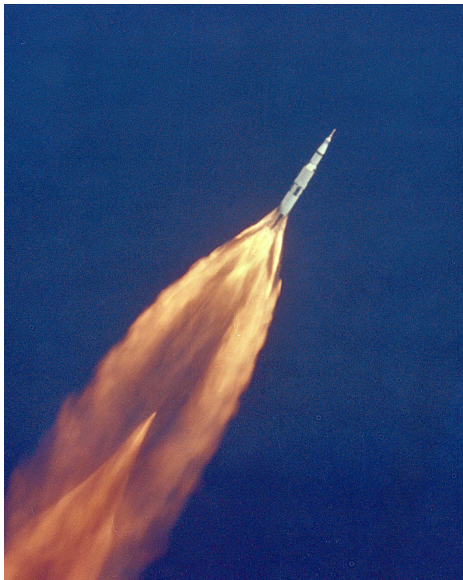
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Outline

1. Setup
2. Filter recursion
3. Filter initialization
4. Constructing the likelihood
5. Practical Considerations

Cool stuff



The basic concept

- named after seminal work of Kálmán (1960) and Kálmán and Bucy (1961)

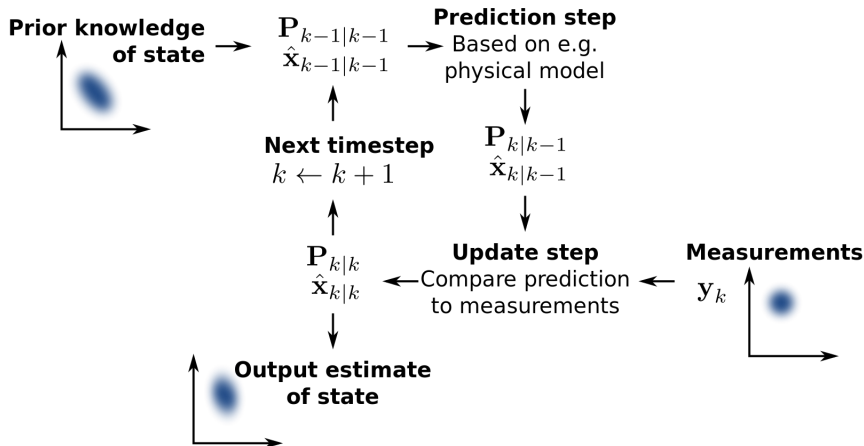


Fig. 1: Source: “Basic concept of Kalman filtering” by Petteri Aimonen

References

- Handout online
- Ljungqvist and Sargent (2012, Chapter 2.5)
- Hamilton (1994, Chapter 13)
- Durbin and Koopman (2012)
- Canova (2007, Chapter 6)

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Gaussian state space

- Solution to linearized RBC model took **state space form**, which can be written as

$$x_{t+1} = Fx_t + w_{t+1}, \quad w_{t+1} \stackrel{iid}{\sim} \mathcal{N}(0, Q) \quad (1)$$

$$y_t = Hx_t + \nu_t, \quad \nu_t \stackrel{iid}{\sim} \mathcal{N}(0, R) \quad (2)$$

- F, H, Q, R are functions of the model parameters θ
- x_t is an $n_x \times 1$ vector of states
- y_t is an $n_y \times 1$ vector of observables
- w_t is a $p \times 1$ vector of structural errors
- ν_t a vector of measurement errors
- Assumption: w_t and ν_t are orthogonal

$$\mathbb{E}_t(w_{t+1}\nu_s) = 0 \quad \forall t+1 \text{ and } s \geq 0$$

Fundamental problem: unobserved states

- This implies that

$$y_t = H(Fx_{t-1} + w_t) + \nu_t \quad (3)$$

- Thus, y_t is normally distributed:

$$y_t \sim \mathcal{N}(HFx_{t-1}, HQH' + R) \quad (4)$$

- If all states were observed, we could directly construct the likelihood $f(y_T, \dots, y_1 | \theta)$
- We could then run optimizer over our estimated parameter set $\tilde{\theta} \subseteq \theta$ to get ML estimate of $\tilde{\theta}$
- Problem: we have **unobserved states** and cannot use equation (4)
- Solution: turn to Kalman filter to back out states from the observed data
→ **Filtering problem**

Initial values for states

- Initial value x_0 distributed as

$$x_0 \sim \mathcal{N}(\hat{x}_{0|-1}, \Sigma_{0|-1}) \quad (5)$$

- subscript -1 denotes information at the beginning of times
- particular observation: y_t
- complete history of observations up to a certain point in time: $y^t = \{y_t, \dots, y_0\}$
- What does this imply for $y_0 = Hx_0 + \nu_0$?

Initial values for observables

- Let's define

$$\hat{y}_{0|-1} \equiv E(y_0|y^{-1}) = E(Hx_0 + \nu_0|y^{-1}) = H\hat{x}_{0|-1} \quad (6)$$

- Mean Squared Error** of y_0 given information up to $t = -1$

$$\begin{aligned} & E \left[(y_0 - \hat{y}_{0|-1}) (y_0 - \hat{y}_{0|-1})' | y^{-1} \right] \\ &= E \left[(Hx_0 + \nu_0 - H\hat{x}_{0|-1}) (Hx_0 + \nu_0 - H\hat{x}_{0|-1})' | y^{-1} \right] \\ &= E \left[H(x_0 - \hat{x}_{0|-1})(x_0 - \hat{x}_{0|-1})' H' + \nu_0 \nu_0' | y^{-1} \right] \\ &= H\Sigma_{0|-1}H' + R \end{aligned} \quad (7)$$

- Hence:

$$y_0 \sim N \left(H\hat{x}_{0|-1}, H\Sigma_{0|-1}H' + R \right) \quad (8)$$

What's the goal?

- Economist only observes the observables: $y^t = \{y_t, \dots, y_0\}$
- Wants to infer the **unobserved state variables** $x_t, \dots, x_0$
- Assumption: Economist knows state space structure, i.e. H, F, R, Q
- Aim: find **recursive** formulas for the state forecast

$$\hat{x}_{t|t-1} \equiv E[x_t | y^{t-1}] \quad (9)$$

and the Mean Squared Error/covariance matrices of the forecast error (FE)

$$\Sigma_{t|t-1} \equiv E \left[\left(x_t - \hat{x}_{t|t-1} \right) \left(x_t - \hat{x}_{t|t-1} \right)' | y^{t-1} \right] \quad (10)$$

- Recursiveness allows for online tracking

Idea

- regression of the (unknown) state FE on observation FE

$$x_t - \hat{x}_{t|t-1} = L_t (y_t - \hat{y}_{t|t-1}) + \eta_t \quad (11)$$

L_t : regression coefficient; η_t : regression residual

- Forecast $\hat{y}_{t|t-1}$ of y_t given by

$$\hat{y}_{t|t-1} = H\hat{x}_{t|t-1} \quad (12)$$

- regression equation can then be written as

$$x_t - \hat{x}_{t|t-1} = L_t (y_t - H\hat{x}_{t|t-1}) + \eta_t \quad (13)$$

- Problem: don't know the left-hand side $\Rightarrow$ can't compute L_t
- But: if we somehow knew L_t , could form a forecast of our state FE

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Computing L_t

- Goal: compute L_t at time t
- General formula for a **population regression**

$$\beta = E(YX') [E(XX')]^{-1}$$

- In our case

$$\begin{aligned} L_t &= E \left[\left(x_t - \hat{x}_{t|t-1} \right) \left(y_t - H \hat{x}_{t|t-1} \right)' \right] \\ &\quad \times \left(E \left[\left(y_t - H \hat{x}_{t|t-1} \right) \left(y_t - H \hat{x}_{t|t-1} \right)' \right] \right)^{-1} \\ &= \Sigma_{t|t-1} H' \left(H \Sigma_{t|t-1} H' + R \right)^{-1} \end{aligned} \tag{14}$$

Stochastic singularity

- Computation requires $\left(H\Sigma_{t|t-1}H' + R\right)^{-1}$
 - For the inversion, state-space model must not feature **stochastic singularity**, i.e. the forecast error matrix of the observables must have full rank
 - Typical requirement: at least as many shocks as observables
 - Having more shocks than observables is not a problem
 - Also requires that there is no collinearity between observables
- Problem if, e.g., all components of budget constraint are observed
- Way out: add measurement error (see Altug 1989; Ireland 2004; Sargent 1989)
 - May also help with model mis-specification (Del Negro and Schorfheide 2009)

Updating step: forecasting x_1

- Rewrite state transition equation

$$\begin{aligned}x_1 &= Fx_0 + w_1 \\&= F\hat{x}_{0|-1} + F\left(x_0 - \hat{x}_{0|-1}\right) + w_1 \\&\stackrel{(13)}{=} F\hat{x}_{0|-1} + F\left(L_0\left(y_0 - H\hat{x}_{0|-1}\right) + \eta_0\right) + w_1\end{aligned}\tag{15}$$

- Forecast tomorrow's state just based on yesterday's forecast and today's observation

$$\hat{x}_{1|0} = E\left[x_1|y^0\right] = F\hat{x}_{0|-1} + \underbrace{FL_0}_{K_0}\left(y_0 - H\hat{x}_{0|-1}\right)$$

- **Kalman Gain**

$$K_0 = FL_0 = F\Sigma_{0|-1}H'\left(H\Sigma_{0|-1}H' + R\right)^{-1}\tag{16}$$

→ how much is state estimate updated based on previous FE

$x_2, x_3, \dots ?$

- We want to use population regression

$$L_1 = \Sigma_{1|0} H' (H \Sigma_{1|0} H' + R)^{-1}$$

- For this, we need covariance/mean squared error matrix $\Sigma_{1|0}$

$$\begin{aligned} \Sigma_{1|0} &= E \left[(x_1 - \hat{x}_{1|0}) (x_1 - \hat{x}_{1|0})' | y^0 \right] \\ &= (F - K_0 H) \Sigma_{0|-1} (F - K_0 H)' + Q + K_0 R K_0' \end{aligned} \quad (17)$$

- From this, we now know that

$$x_1 \sim \mathcal{N}(\hat{x}_{1|0}, \Sigma_{1|0}) \quad (18)$$

- We can now start over again, delivering the recursion we were looking for

Summary

At time t , given $\hat{x}_{t|t-1}$, $\Sigma_{t|t-1}$ and observing y_t

1. Compute the forecast error in the observations using

$$a_t = y_t - H\hat{x}_{t|t-1} \quad (19)$$

2. Compute the **Kalman Gain** K_t using

$$K_t = F\Sigma_{t|t-1}H' \left(H\Sigma_{t|t-1}H' + R \right)^{-1} \quad (20)$$

3. Compute the state forecast for next period given today's information

$$\hat{x}_{t+1|t} = F\hat{x}_{t|t-1} + K_t \left(y_t - H\hat{x}_{t|t-1} \right) = F\hat{x}_{t|t-1} + K_t a_t \quad (21)$$

4. Update the covariance matrix

$$\Sigma_{t+1|t} = (F - K_t H) \Sigma_{t|t-1} (F - K_t H)' + Q + K_t R K_t' \quad (22)$$

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Initialization

- How to initialize filter at $t = 0$ where no observations are available?
→ start with **unconditional** mean $E(x)$ and Variance Σ
- Given covariance stationarity, the unconditional mean is

$$E(x) = Ex_{t+1} = E(Fx_t + w_{t+1}) = FE(x) \Rightarrow (I - F)E(x) = 0$$

hence, $E(x) = 0$

- For the covariance matrix, we have

$$\begin{aligned}\Sigma &= E \left[(Fx_t + w_t) (Fx_t + w_t)' \right] \\ &= E \left[Fx_t x_t' F' + w_t w_t' \right] \\ &= F \Sigma F' + Q\end{aligned}\tag{23}$$

→ so-called **Lyapunov-equation**

Initialization: the slow solution

- If A, B, C are conformable matrices

$$\text{vec}(ABC) = (C' \otimes A)\text{vec}(B)$$

- Hence

$$\text{vec}(\Sigma) = (F \otimes F)\text{vec}(\Sigma) + \text{vec}(Q)$$

- This has solution

$$\text{vec}(\Sigma) = \left(I_{n_x^2} - (F \otimes F) \right)^{-1} \text{vec}(Q) \quad (24)$$

- Problem: Inversion of n_x^2 matrix $\rightarrow$ sloooooow

Lyapunov solvers in Dynare

- Dynare offers a menu of superior alternatives via the

```
lyapunov = OPTION
```

- The `lyapunov=default` option uses the Schur decomposition-based Bartels-Stewart algorithm
- The `lyapunov=doubling` option uses the doubling algorithm, which is usually a lot faster for large models
- For big models, the `lyapunov=square_root_solver` of MATLAB's Control System Toolbox may also work well

Initialization: Diffuse Kalman filter

- Consider model with unit root: does not violate Blanchard and Kahn (1980) conditions
 - Problem: unconditional variance does not exist for some variables, only the conditional one
- cannot use unconditional one for initialization
- Alternative: **diffuse Kalman filter** (De Jong 1991; Koopman and Durbin 2003) that considers initial state as diffuse vector (let variance go to infinity)
 - requires specifying the `diffuse_filter`-option of Dynare (also important for treating unit roots as being stable (`options_.qz_criterium`))

Diffuse initialization of the filter

- Vector x is split in $\begin{bmatrix} x^1 & x^2 \end{bmatrix}$, where x^1 contains non-stationary variables and x^2 stationary ones
- Koopman and Durbin (2003) compute second order expansion of the Kalman filter formulas when the variance of the non-stationary variables tends toward infinity
- Initial values for the state variables are $x_{0|-1}^1 = x_{0|-1}^2 = 0$
→ unconditional mean of the stationary elements in x_t , with no effect for non-stationary ones

Diffuse filter: initial co-variance matrix

- The initial co-variance matrix is

$$P_{0|-1} = P_{0|-1}^{\infty} + P_{0|-1}^* = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & \Sigma_{x_2} \end{bmatrix}, \quad (25)$$

where

- the identity matrix I of the same dimensions as F_{11} is the diffuse prior on the initial values of the stochastic trends
- Σ_{x_2} is the co-variance matrix of the stationary part of x_t , i.e. x_t^2
- Schur decomposition can be used to find the actual number of unit roots
→ important if model has co-integrated variables
- While $P_{t|t-1}^{\infty}$ is different from zero, the filter is in a diffuse step

Exiting the diffuse filter

- For non-degenerate models that are informative about the state of the world, the number of diffuse steps d is smaller than t
→ algorithm falls back on standard recursions where $x_{d|d-1}, P_{d|d-1}$ can be used for initialization
- However, diffuse filter would never exit with only stationary observables
→ if non-stationary states are not projected onto some observation, they provide no information about state location
- Dynare automatically tests if there exist unit roots not projected onto any observable
→ set entries of $P_{0|-1}^{\infty}$ associated to such state automatically to 0
- If only unobserved non-stationary states: $P_{0|-1}^{\infty}$ has rank zero
→ use standard Kalman filter using $P_{0|-1}^*$
- Typical example: inflation rates are not informative about price level in model

Problem with co-integrated variables and improper prior

- The diffuse prior is improper: no direct comparison of marginal data density of different models is possible
→ Special care (i.e. presample) must be used for models comparison
- Dynare internally works with non-stationary trends instead of non-stationary variables
- Core problem otherwise: when x_t^1 contains co-integrated variables, the limit co-variance matrix of the diffuse state vector is not given by $\lim_{k \rightarrow \infty} kI$
→ diffuse prior on all involved states would imply too much volatility of initial conditions (efficiency of filter/smoothen drastically deteriorates)
- Dynare therefore employs a Schur decomposition to isolate trends corresponding to different unit roots
→ work directly with stochastic trends instead of trending variables by rotating the variable space
- Requires resorting to univariate diffuse Kalman filter recursions (Koopman and Durbin 2000)

Univariate Kalman filter

- The updating step using the Kalman gain involves inversion of potentially big matrix

$$K_t = F\Sigma_{t|t-1}H' \left(H\Sigma_{t|t-1}H' + R \right)^{-1} \quad (20)$$

- Koopman and Durbin (2000): update state prediction sequentially using one observed series at a time instead of using the full vector of observations
→ avoids matrix inversion
- Univariate Kalman filter
 - can provide decent speed-up for large models
 - works even if there is stochastic singularity
- However: stochastic singularity typically indicates a problem with the model that cannot simply be solved with a different filter!

Fast Kalman filter

- Most expensive computation involves updating the covariance matrix

$$\Sigma_{t+1|t} = (F - K_t H) \Sigma_{t|t-1} (F - K_t H)' + Q + K_t R K_t' \quad (22)$$

due to first part involving multiplication of three $n_x \times n_x$ matrices: $\mathcal{O}(n_x^3)$

- Central insight: $\Sigma_{t+1|t}$ is rarely full rank
- Idea: use a rank reduction technique that considers differences in state variances

$$\Delta \Sigma_{t|t-1} = W_t M_t W_t', \quad (26)$$

where $\text{rank}(M_t)$ is at most $\min(n_x, n_y)$

→ allows large speed-up if model has many more states than observables (Smets and Wouters (2007) has 50 states but only 7 observables)

Chandrasekhar recursion

- Herbst (2015) suggests the following implementation of Chandrasekhar recursion

$$G_t = G_{t-1} + HW_{t-1}M_{t-1}W'_{t-1}H' \quad (27)$$

$$K_t = K_{t-1} + FW_{t-1}M_{t-1}W'_{t-1}H' \quad (28)$$

$$W_t = (F - K_tG_tH)W_{t-1} \quad (29)$$

$$M_t = M_{t-1} + M_{t-1}W'_{t-1}H'G_{t-1}^{-1}HW_{t-1}M_{t-1} \quad (30)$$

with initialization

$$G_1 = H\Sigma_0H' + RQR \quad (31)$$

$$K_1 = F\Sigma_0H' \quad (32)$$

$$W_1 = K_1 \quad (33)$$

$$M_1 = -G_1^{-1} \quad (34)$$

- Recursion only works if Σ_0 exists
→ not suitable for models with unit roots

Steady State Kalman filter

- It can be shown that the matrix recursions in the Kalman filter converge (See Hamilton 1994, Chapter 13.5)
 - Kalman gain and forecast error variance matrix will settle on final values
 - Continuing with recursions after they have settled within reasonable tolerance is computationally expensive
- check for convergence and then continue with fixed matrices
- Dynare by default uses this **steady state Kalman filter**

Heteroskedastic filter

- Our description used time-invariant matrices state space matrices H, F, Q, R , but Kalman filter can accommodate time-varying ones
- Somewhat problematic from the perspective of rational expectations modeling as implied structural breaks are essentially MIT shocks
- Dynare has `heteroskedastic_filter` option to allow heteroskedasticity via Q_t

```
heteroskedastic_shocks;  
var e1;  
periods 86:87, 89:97;  
scales 0.5, 0;  
var e1;  
periods 88;  
values 0.1;
```

- Can e.g. be used for increasing variance in Covid period

Options in Dynare

- `kalman_algo = INTEGER`
 - 0 automatically use Multivariate Kalman Filter for stationary models and Multivariate Diffuse Kalman Filter for non-stationary models
 - 1 Use Multivariate Kalman Filter
 - 2 Use Univariate Kalman Filter
 - 3 Use Multivariate Diffuse Kalman Filter
 - 4 Use Univariate Diffuse Kalman Filter
- `lik_init = INTEGER`
 - 1 Use unconditional variance
 - 3 Use diffuse initialization
 - better to use `diffuse_filter`-option
- `fast_kalman_filter`: use Chandrasekhar recursions

Kalman smoothing and other outputs

- The Kalman filter recursions provide us with **filtered variables** $\mathbb{E}_t x_{t+1}$, i.e. the best prediction for tomorrow's state given information up to today
- We can also compute **updated variables** $\mathbb{E}_t x_t$, i.e. our best estimate of the state today given information up to today (remember that x_t is not in the information set here!)
- Note: the terminology here follows Dynare as there seems to be no standard for naming these two objects
- More importantly, we are regularly interested in our best estimates of shocks and states given the full observed data up to time T
- The **Kalman Smoother** provides recursions for obtaining these estimates $\mathbb{E}_T(x_t)$ and $\mathbb{E}_T(w_t)$ by working backwards in time (for details, see the handout)

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From filtering to estimation

- Typically we are not interested in filtering per se, but rather in estimating a model
- Start with complete history of observables up to time t : $y^t = \{y_t, \dots, y_0\}$
- Likelihood function $f(y_T, \dots, y_0)$ can be factored as

$$f(y_T, \dots, y_0) = f(y_T | y^{T-1}) \times f(y_{T-1} | y^{T-2}) \times \dots \times f(y_1 | y^0) \times f(y_0)$$

- Earlier we derived

$$y_t \sim \mathcal{N} \left(H \hat{x}_{t|t-1}, \underbrace{H \Sigma_{t|t-1} H' + R}_{\equiv \Omega_t} \right)$$

Log-likelihood

- Hence, the probability density of observing y_t given y^{t-1} is given by

$$f(y_t|y^{t-1}) = \frac{1}{\sqrt{(2\pi)^{n_y} \det(\Omega_t)}} e^{-\frac{1}{2}(y_t - H\hat{x}_{t|t-1})' \Omega_t^{-1} (y_t - H\hat{x}_{t|t-1})},$$

- Taking logs leads to the **log-likelihood function** for each observation:

$$\log(f(y_t|y^{t-1})) = -\frac{n_y}{2} \log(2\pi) - \frac{1}{2} \log(\det(\Omega_t)) - \frac{1}{2} a_t' \Omega_t^{-1} a_t \quad (35)$$

→ Kalman filter delivers everything needed to compute the likelihood

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Finding the global maximum

- Finding a **global maximum** is hard!
- Newton-type optimizers are inherently **local**
- One way out: try numerous starting points
- Alternative: use **global optimizers** like simulated annealing (**Coronaetal1987**; e.g. Goffe et al. 1994) or covariance matrix adaptive evolutionary strategy (CMA-ES) (Hansen et al. 2003)
- Particularly CMA-ES (mode_compute=9 in Dynare) seems to perform well in practice (Andreasen 2010)
- Dynare supports various different optimizers, calling them sequentially seems to work pretty well in hard cases
→ Dynare 6.x has option `additional_optimizer_steps` to trigger mode finders in sequence

Mode-finders in Dynare

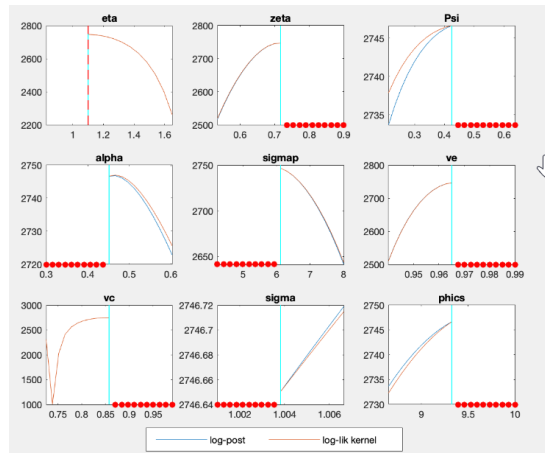
- Dynare offers various types of mode-finders via `mode_compute` option
- Essentially three classes of optimizers:
 1. Local mode-finders (typically Newton-type):
 - Matlab's `fmincon`, `fminunc`: `mode_compute=1,3`
→ require Optimization Toolbox
 - Dynare's implementation of `csminwel`, `newrat`: `mode_compute=4,5`
 2. Global mode-finders:
 - Variants of simulated annealing: `mode_compute=2,102`
 - Variants of simplex algorithm: `mode_compute=7,8`
 - Combination of the two: `simpsa`: `mode_compute=10`
 - CMA-ES (Covariance Matrix Adaptation Evolution Strategy): `mode_compute=9`
 - Particle Swarm: `particleswarm` `mode_compute=12`
→ requires Global Optimization Toolbox
 3. Monte Carlo-based optimization routine: `mode_compute=6`
- Dynare's interface also allows employing user-defined algorithms

Mode-finding: some considerations

- Users face trade-off between speed advantage of local optimizers and getting stuck at local mode
- Global optimizers tend to stop at points where Hessian is not positive definite (as they typically do not rely on gradients)
→ may be preferable to run local optimizer after global one
- Historically, `mode_compute=4` has been the default in Dynare, but `mode_compute=5` regularly outperforms it and is the new default in 6.x
- `analytic_derivation` allows using analytically computed gradient and Hessian at the mode instead of numerically computed one
→ supported by `mode_compute=1,3,4,5,101` for stationary models without missing observations (`kalman_algo<3`)

Diagnostics: mode check plots

- The `mode_check`-option provides an important diagnostics for checking issues:
 - Corner solutions due to hitting the explicit (or implicit) prior bound
 - Identification issues (flat likelihood)
 - Complete failure of optimization
 - Implausibly high values for the likelihood
- Problems in the model setup or the mapping of the data to the model variables can cause such issues
→ first check for such issues
- Nevertheless, non-positive definite Hessians at the final point regularly occur



Optimality

- Kalman filter is basically a least squares estimator
 - If initial conditions and innovations are normal, it is the best predictor! You cannot get better!
 - Otherwise, it is the best linear predictor
 - Can only be used to construct likelihood function for linear (asymptotically) Gaussian state-space systems
- works perfectly for first-order approximated DSGE models
- does not work with higher order approximations/non-linearities

Missing observations and mixed frequency

- Nothing in the Kalman filter recursions requires the involved matrices to be the same in every iteration
- Missing observations can thus be easily handled by adjusting the matrices using a selector matrix
→ Dynare does this automatically
- Allows also mixed frequency estimation:
 - set up model in the highest frequency
 - set variables not available at this frequency to missing
 - define a mapping from the quarterly flow variables in the data to the monthly ones in the model

Making estimation faster

- Diffuse filter is slower than basic filter: you should weight the benefit of using a non-stationary model
- Using a doubling algorithm to compute the unconditional variance is often faster: use estimation option `lyapunov=doubling`
- For stationary models: use estimation option `fast_kalman_filter`
- If you can provide Dynare with an explicit steady state for your model with `steady_state_m` block
- Use model option `use_d11` for medium or large size models
- The benefit of the above options is highly model dependent. Tests are required to find the best choices for any specific model.

Observation equations

- Up to this point, we have been silent on how to map the observed data to the model variables
- This is the topic of specifying the **observation equations**
- The biggest problem is stationarity in the model vs. trends in the data
- The guiding principle is: **the data variable needs to perfectly correspond to a variable defined in the model**
- A lengthy treatment of many possible cases is available in Pfeifer (2013)
- We will most of the time consider demeaned growth rates of variables together with a loglinear model
- This gives rise to a straightforward observation equation of the form:
 $y_{\text{obs}} = y - y(-1)$;

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